

KEERTHISHREE T S

SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Final-year B.E. Computer Science Engineering student (CGPA 8.91) focused on backend and distributed systems. Builds software from the storage layer up — a fault-tolerant job queue on PostgreSQL, an OpenTelemetry trace collector, a real-time streaming service — and backs every claim with tests, failure injection and measured benchmarks. Also ships full-stack products with Next.js and TypeScript. Strong in data structures and algorithms, and seeking a software engineering role building reliable, scalable systems.

EDUCATION

SNS College of Technology, Coimbatore

2023 – 2027

B.E., Computer Science Engineering — CGPA: **8.91 / 10**

MNC School

2023

Higher Secondary Education — Score: 83/100

EXPERIENCE

AI & ML Intern — Microsoft Elevate

Jan 2025 – Feb 2025

- Built a fake-news detection pipeline in Python: NLP text preprocessing, TF-IDF feature extraction and a Logistic Regression classifier, validated on accuracy, precision, recall and a confusion matrix.
- Shipped a Streamlit web interface that classifies user-submitted article text in real time.

PROJECTS

Tether — Durable Distributed Job Queue

[GitHub](#)

Python, PostgreSQL 16, Docker, Prometheus, GitHub Actions CI

- Designed a job queue with **at-least-once delivery** on PostgreSQL: workers claim tasks with **SKIP LOCKED leases**, a sweeper re-queues expired leases, and **fencing tokens** stop a paused worker from acknowledging work already handed to another.
- Added idempotency keys committed in the same transaction as the acknowledgement, jittered exponential backoff, priority with starvation boost, and a dead-letter queue, so a task is never silently dropped.
- Sustained **3,245 tasks/s** with p50 latency of 50.5 ms (4 worker processes, benchmarked on one machine); a task on a **SIGKILL**ed worker is redelivered in **1,985 ms** under a 2 s lease.
- Proved the guarantees with a **chaos test suite** (worker SIGKILL/SIGSTOP, Postgres restarts) among **73 tests**, and built a live dashboard with Prometheus metrics.

Wake — OpenTelemetry Distributed Trace Collector

[GitHub](#)

Python, OTLP/HTTP, Protocol Buffers, SQLite

- Built a collector that ingests spans over OTLP, stitches them into traces across services, repairs clock skew and orphaned spans, and serves flame charts; wrote a dependency-free Protobuf wire decoder (**34,881 spans/s**).
- Profiled storage as the bottleneck and raised write throughput **3.9x (1,941 → 7,485 traces/s)** by batching transactions and storing one row per trace; covered by **108 tests**.

Headway — Real-Time Transit Reliability Monitor

[GitHub](#)

Python, FastAPI, WebSockets, Protobuf, Next.js, React, TypeScript

- Streamed live GTFS-Realtime vehicle feeds to browsers over **WebSockets**, detecting bus bunching, ghost vehicles and fleet anomalies; cut per-tick payload **113.7 KB → 14.6 KB** with server-side viewport filtering. **81 tests**.

Full-Stack Blog Platform

[GitHub](#)

Next.js 16, React, TypeScript, Auth.js v5, PostgreSQL, Prisma, Vercel

- Built a multi-user publishing app with React Server Components, email/password and **Google OAuth** login (bcrypt hashing), a PostgreSQL schema on Prisma, and CI/CD to Vercel that runs database migrations on deploy.

Also built

- Keel** — programming language with a bytecode compiler, stack VM and mark-and-sweep GC, up to 3x faster than its tree-walking interpreter (204 tests). **Berth** — async HTTP reverse proxy with consistent hashing, circuit breakers and health checks, benchmarked against nginx. **Anchor** — append-only key-value store with crash recovery.

TECHNICAL SKILLS

Languages

Python, TypeScript, JavaScript, SQL, C++, HTML, CSS

Backend & Systems

FastAPI, REST APIs, WebSockets, asyncio, concurrency, distributed systems (leases, idempotency, retries, circuit breakers, consistent hashing), HTTP/1.1, Protocol Buffers, OpenTelemetry, JWT / OAuth 2.0

Databases

PostgreSQL, MySQL, SQLite, Prisma ORM, SQLAlchemy, schema design, transactions and indexing

Frontend

React, Next.js (App Router, Server Components), Tailwind CSS

Testing & Ops

Pytest, integration and chaos testing, benchmarking and profiling, Docker, Git / GitHub, GitHub Actions CI/CD, Prometheus, Linux, Vercel, Render

AI / ML

scikit-learn, Pandas, NumPy, NLP (TF-IDF), Streamlit, LLM APIs

CS Foundations

Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks